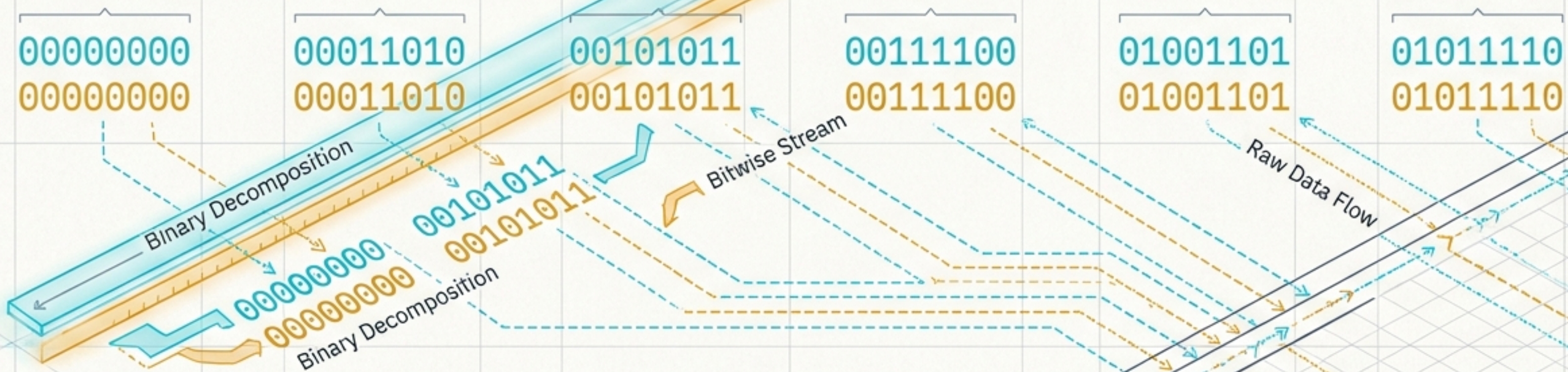


Beyond the Hardware: Decoding Layer 2 Identifiers

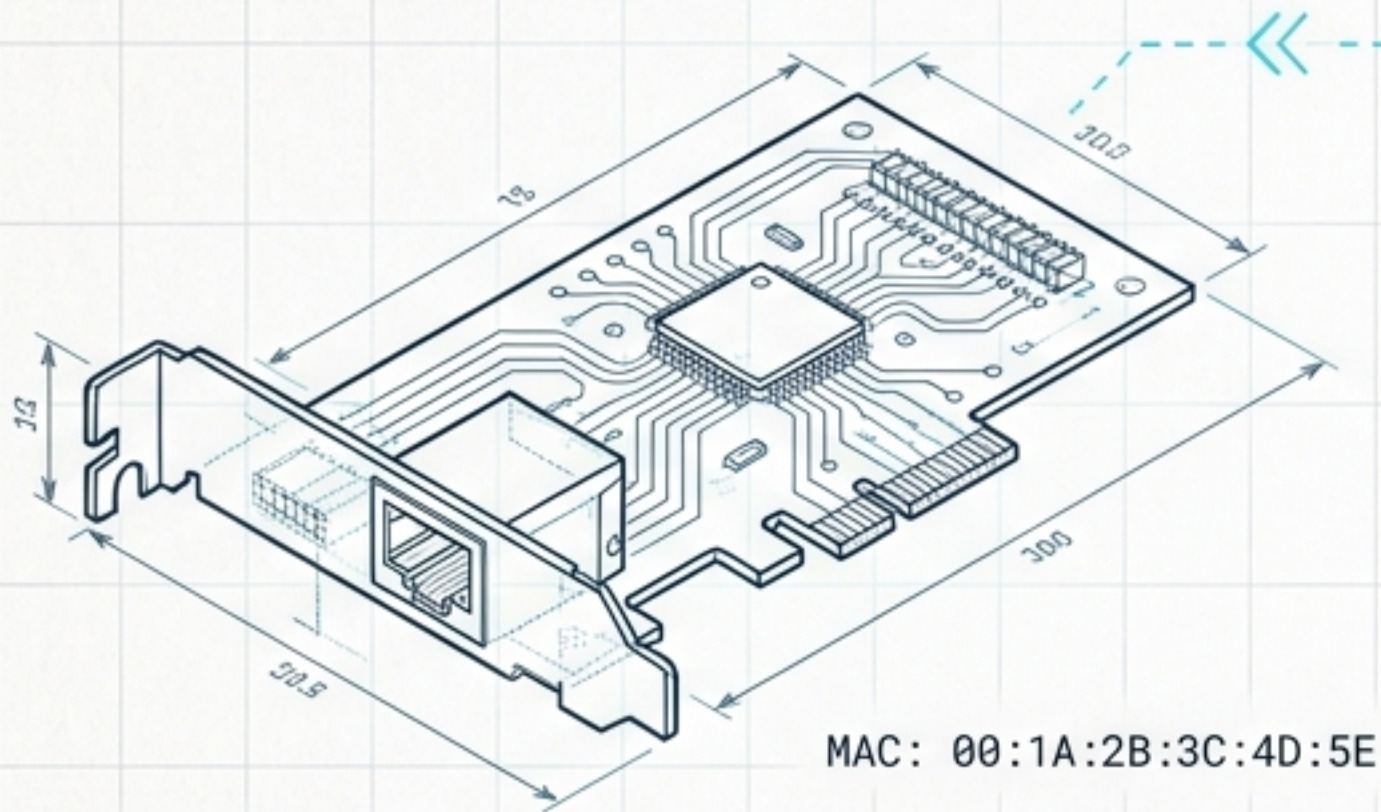
An architectural guide to the anatomy, mechanics,
and security of MAC addresses.

00:1A:2B:3C:4D:5E



The Core Premise of Layer 2

What it is: A Layer 2 identifier used for local Ethernet communication.



The Crucial Misconception

If I know your MAC address, I know who you are.



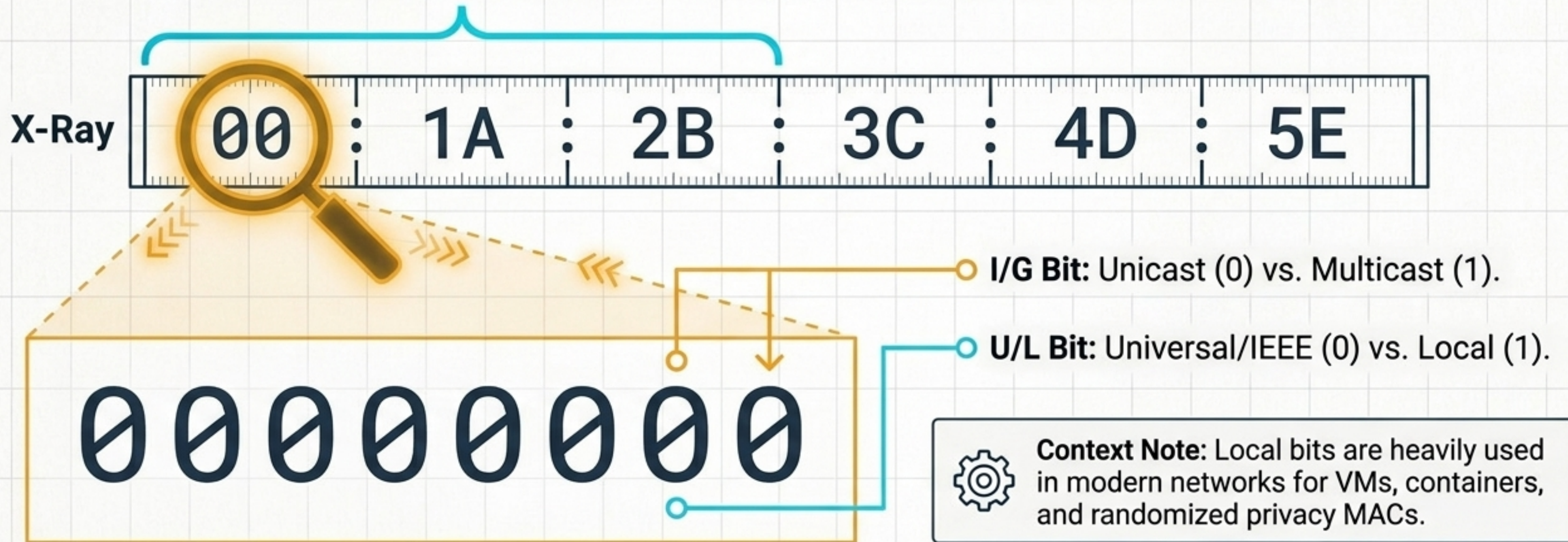
The Reality

A MAC address identifies a network interface, not a person, device owner, or authenticated user. It can be dynamically randomized, virtualized, and spoofed by the operating system.

The 48-Bit Anatomy (EUI-48)

48 bits = 6 bytes = 12 hex digits (1 hex digit = 4 bits).

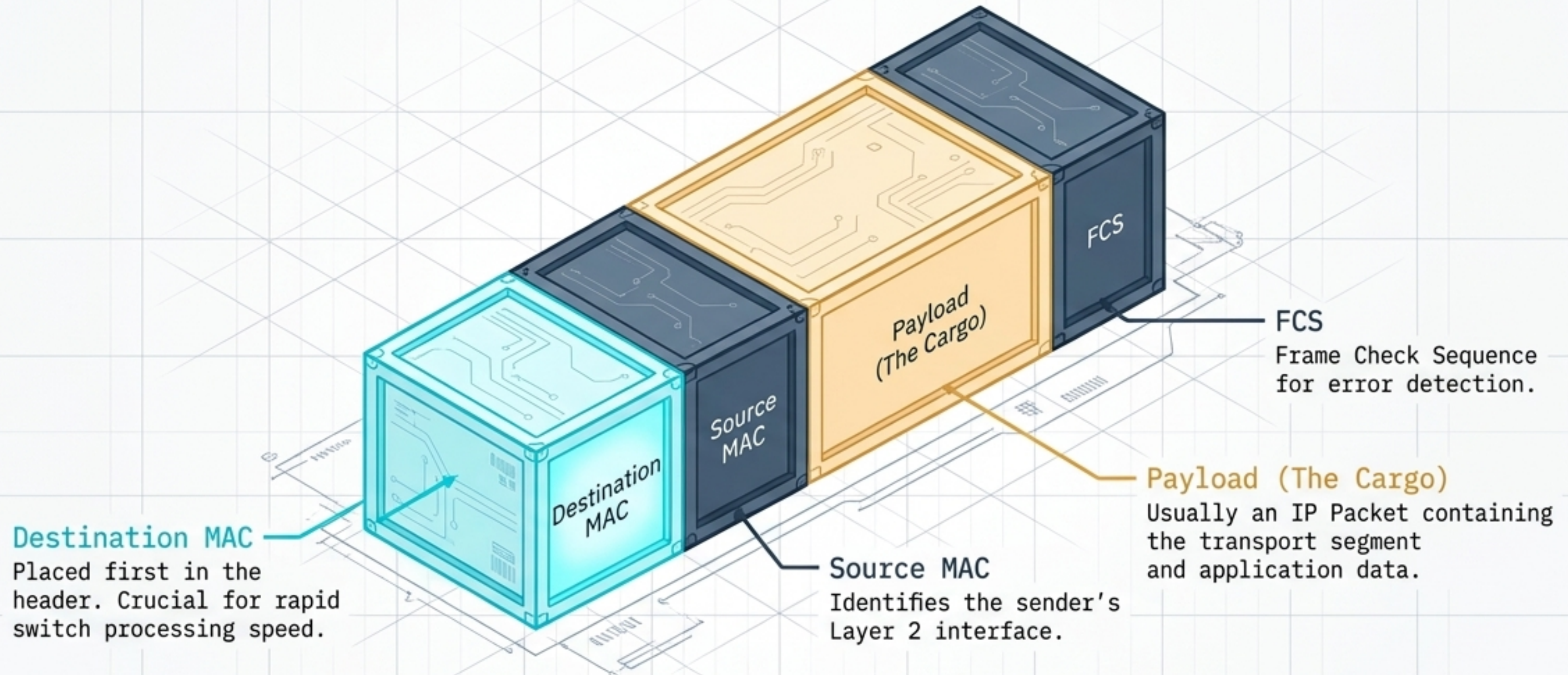
OUI (First 24 bits): Organizationally Unique Identifier. An IEEE-allocated block.
Note: Identifies the allocated block, not necessarily the physical manufacturer.



The Diagnostic Matrix: MAC vs. IP Addresses

	MAC Address	IP Address
Layer	Layer 2 (Data Link)	Layer 3 (Network)
Format	48-bit Hexadecimal	32-bit (IPv4) / 128-bit (IPv6)
Primary Purpose	Local delivery on the current segment	End-to-end logical routing
Router Behavior	Replaced per link at every hop	Remains persistent across boundaries
Hardware Usage	Used by switches for forwarding decisions	Used by routers for pathing decisions

Inside the Ethernet Frame Container

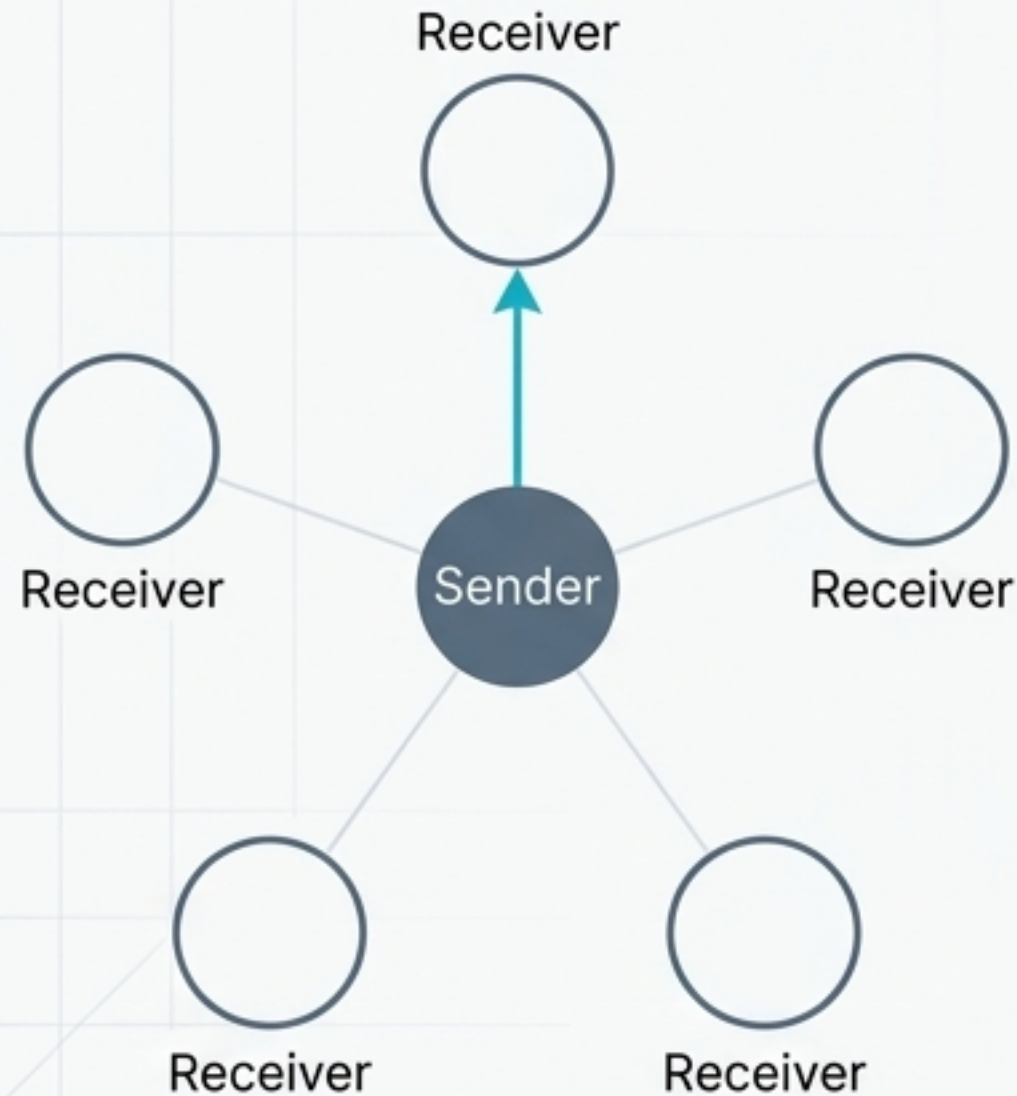


Operational Takeaway: The switch only cares about the outer Layer 2 envelope to make its move; it does not read the inner payload.

The Three Ethernet Delivery Scopes

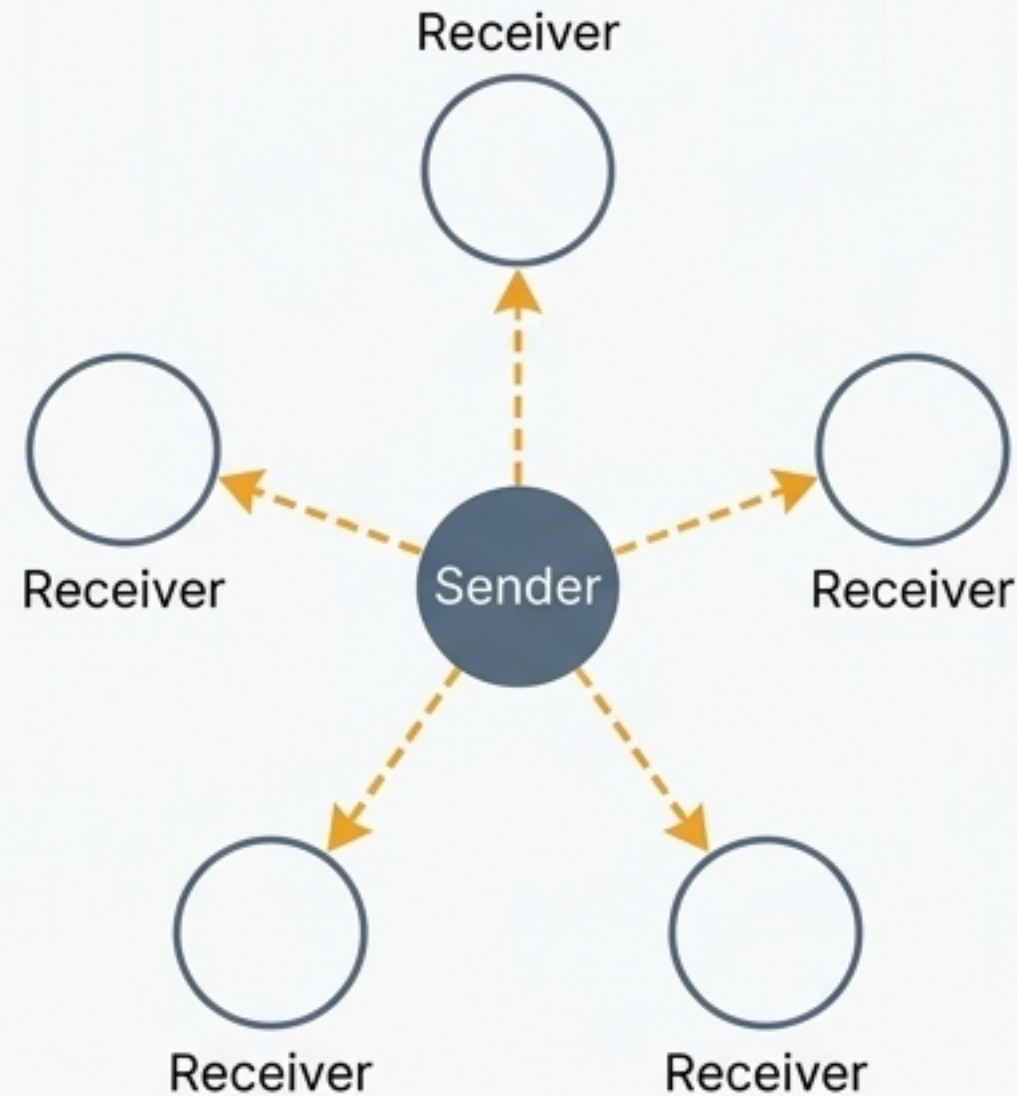
Unicast (One-to-One)

Destination is a specific MAC
(e.g., AA:BB...).



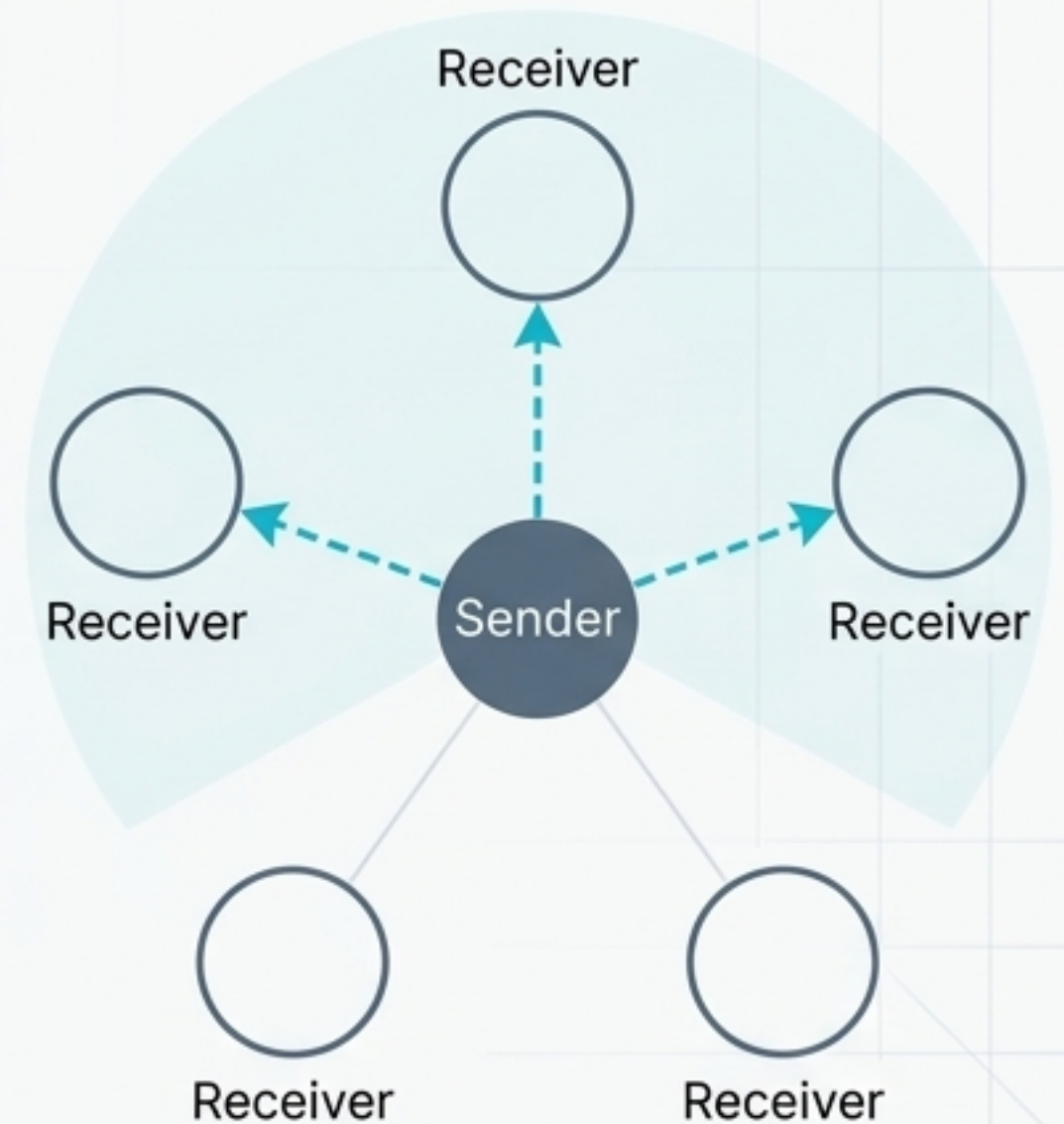
Broadcast (One-to-All)

Destination is FF:FF:FF:FF:FF:FF.

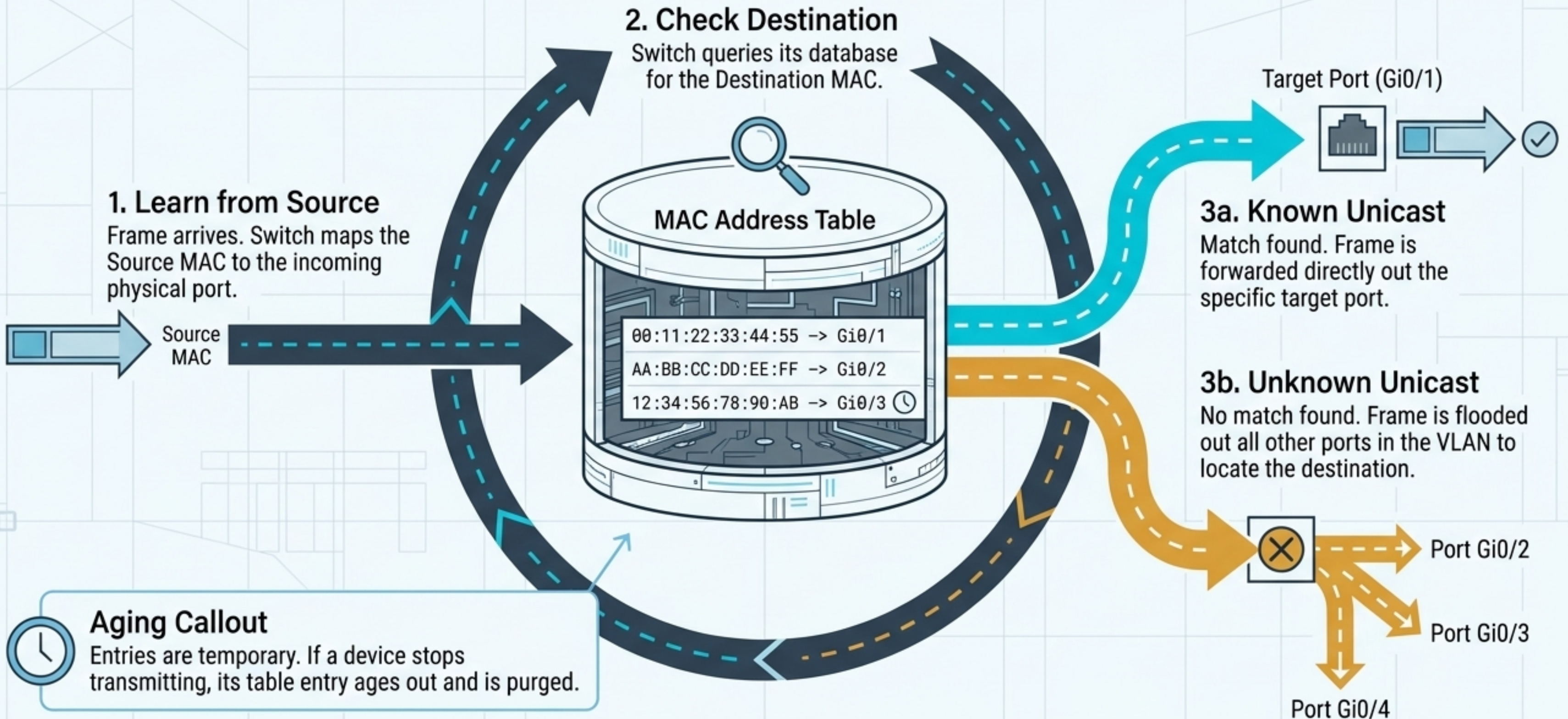


Multicast (One-to-Many)

Destination has the I/G bit set to 1.

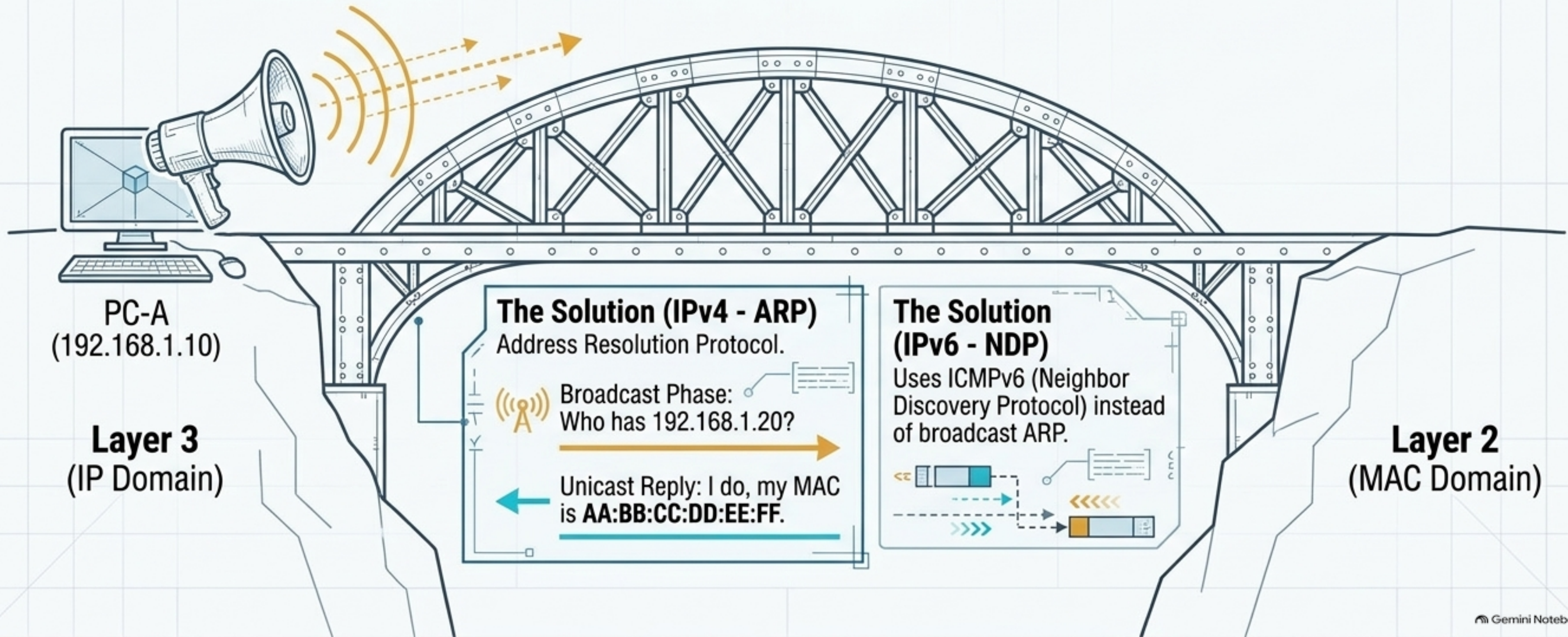


The Mechanics of Switching: Learn and Forward

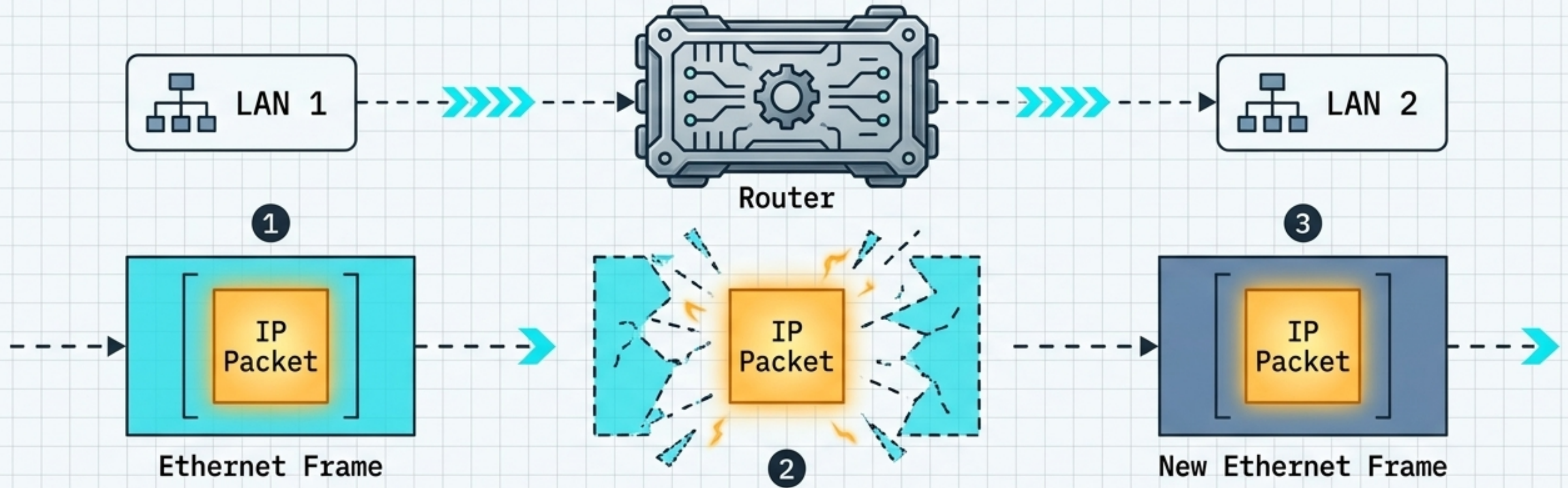


The Missing Link: ARP and NDP

The Problem: I know the destination IP, but not their MAC address. How do I build the Ethernet frame?



The Layer 2 Boundary: The Router Rewrite



Same Subnet Traffic:

- Host builds frame addressed directly to the Destination Host's MAC.

Different Subnet Traffic:

- Host builds frame addressed to the Default Gateway MAC.

The Rewrite Rule:

Routers terminate the incoming Layer 2 frame and generate a **completely new Ethernet frame** for the outgoing interface. MA:BB:CC:DD:EE:FF. **MAC addresses** are strictly hop-by-hop.

The Security Illusion: Identifier $\neq$ Authenticator

00:1A:2B:3C:4D:5E

The Fundamental Flaw:

A switch's MAC table only answers the question: Where did I learn this address?

The Missing Proof:

It NEVER answers:
Has this device proven it is authorized?

The Shift:

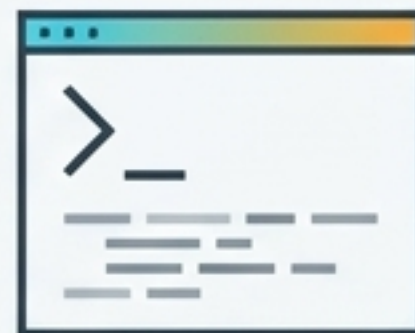
Because a MAC address is just an observable tag, relying on it as a security boundary is inherently dangerous.

MAC Spoofing Demystified



1. Observe

An attacker sniffs the network to identify an authorized MAC address.



2. Clone

The attacker alters their OS configuration to present the cloned Layer 2 identity.



3. Bypass

The attacker evades weak MAC-based access controls.

The Crucial Limit: Spoofing only changes the name tag. It **does not** grant the attacker the victim's passwords, private keys, or encrypted sessions.

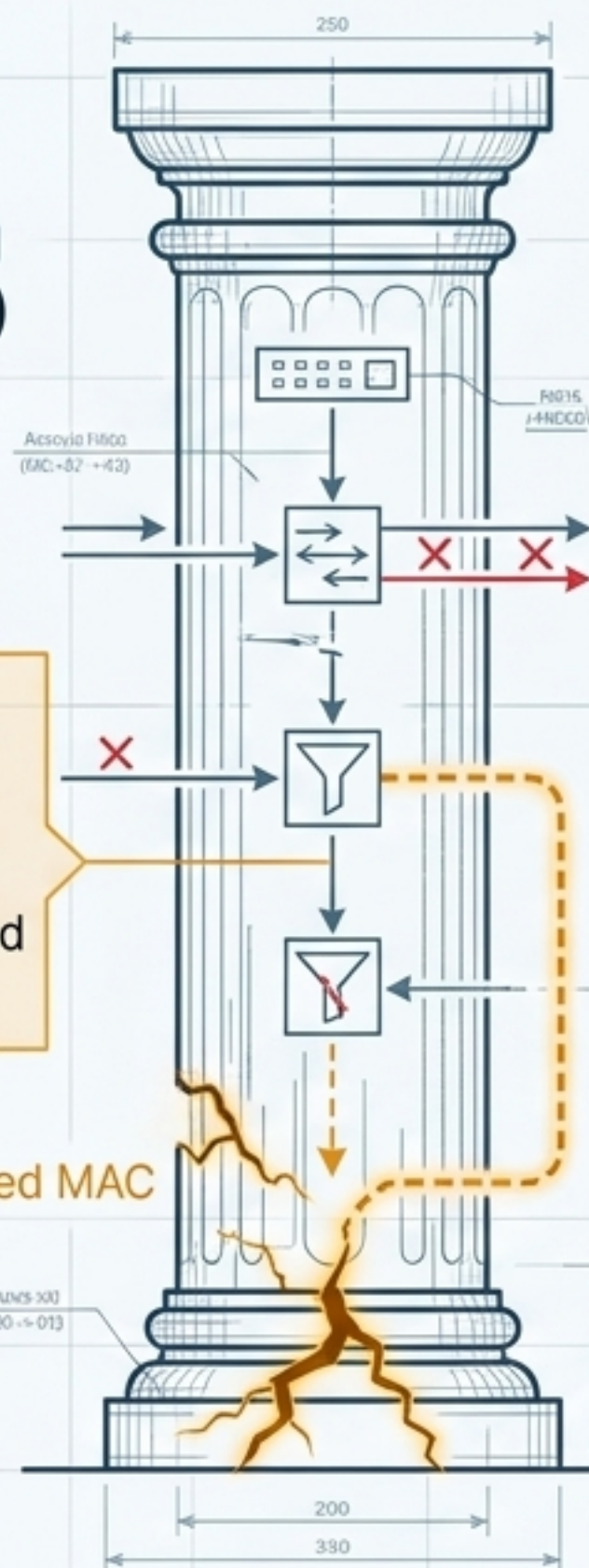
Evaluating Legacy Layer 2 Defenses

Pillar 1: MAC Filtering (Allowlists/Denylists)

Mechanism:
Switch permits only configured MACs.

Verdict:
Weak Security. Easily bypassed by observing and copying allowed addresses. Good for basic inventory, bad for access control.

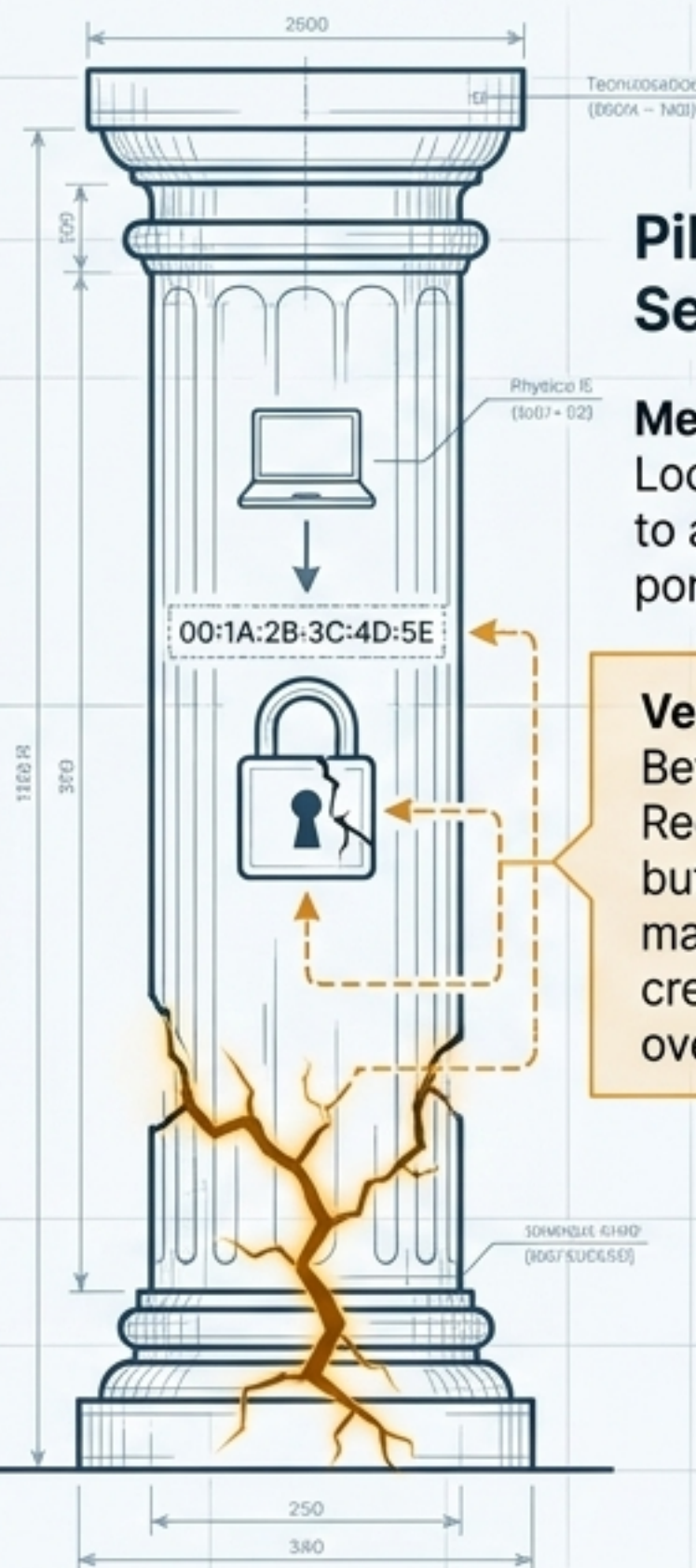
Spoofed MAC



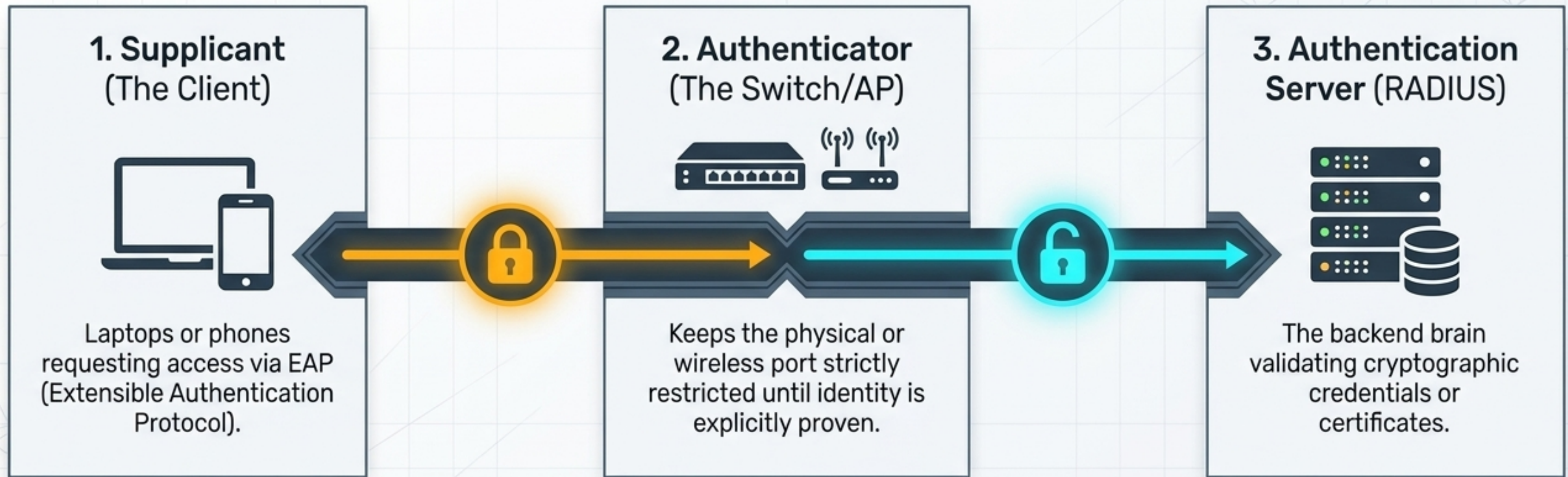
Pillar 2: Switch Port Security

Mechanism:
Locks a specific MAC address to a specific physical switch port.

Verdict:
Better, but Flawed. Reduces mobility of attacks but still relies entirely on a manipulable identifier and creates high management overhead.

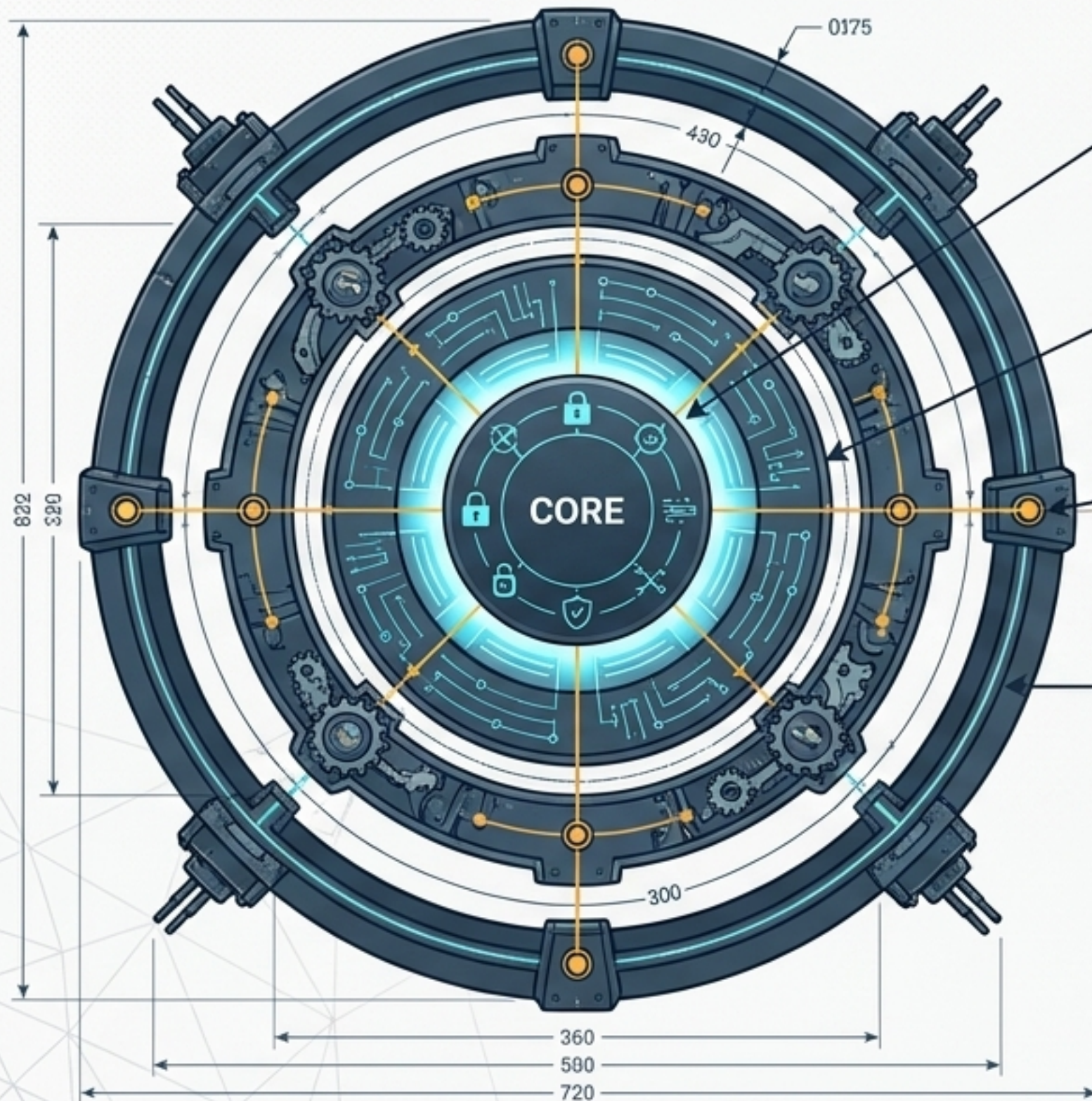


802.1X: True Port-Based Access Control



The Paradigm Shift: Requires cryptographic proof of identity before normal network access is granted, rendering MAC spoofing ineffective.

The Defense in Depth Architecture



Core Ring: 802.1X
(Cryptographic Network Admission)

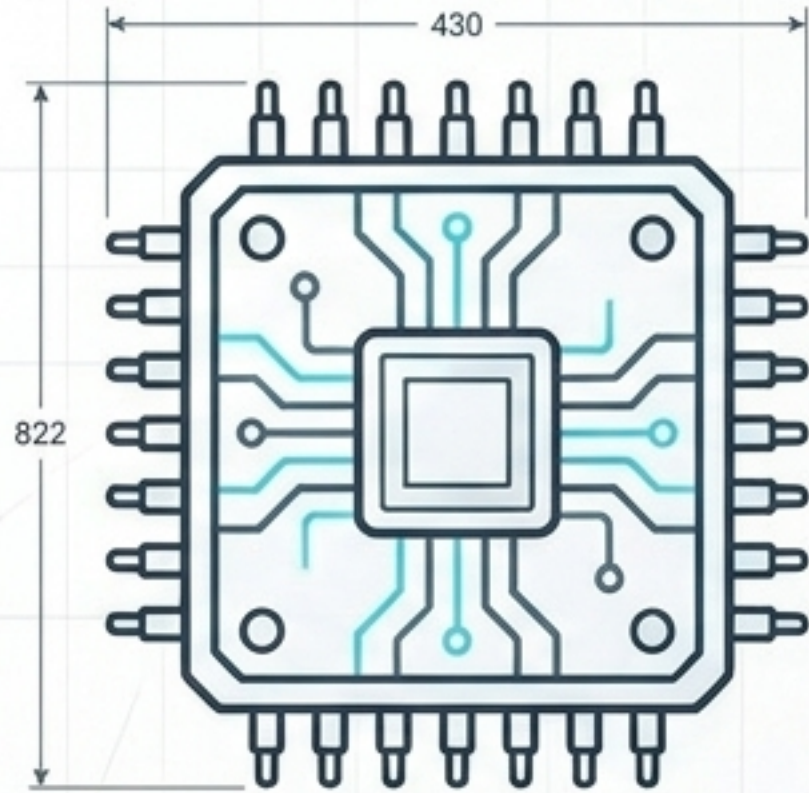
Ring 2: VLANs
(Layer 2 Segmentation and broadcast isolation)

Ring 3: Port Security
(Restricting unexpected MAC anomalies and loops)

Outer Ring: Layer 3 ACLs & Endpoint Security
(IP-level control and device protection)

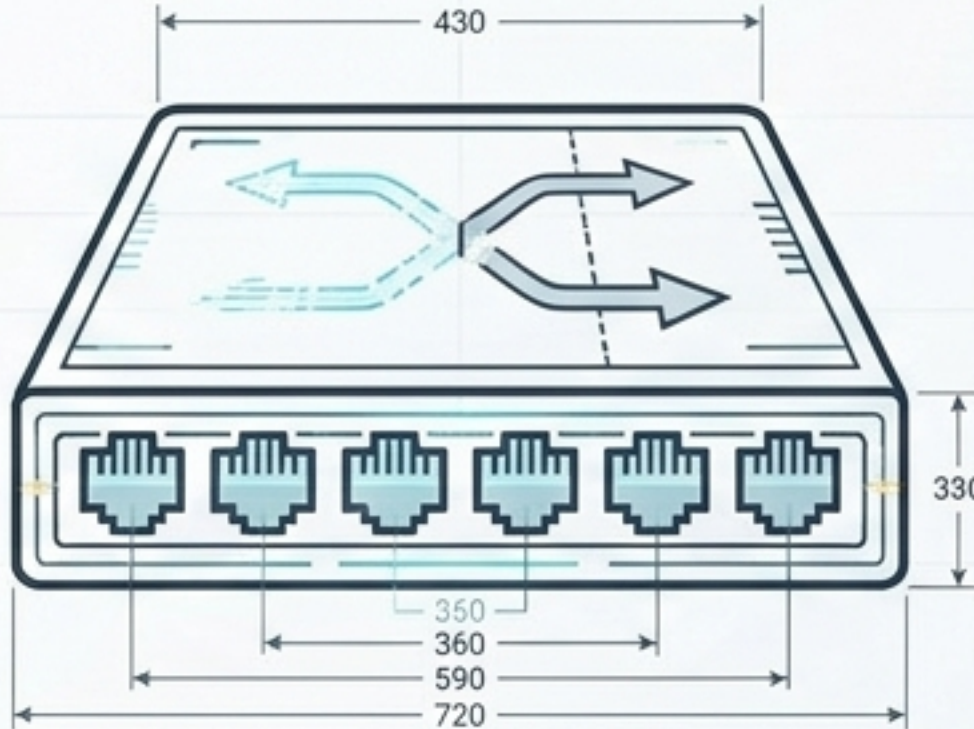
The Rule of Mature Networks:
Assume MAC addresses will be spoofed; rely on overlapping, verifiable controls.

The Layer 2 Blueprint Summary



Anatomy (Form)

48-bit (EUI-48) local identifiers containing OUI, Universal/Local, and Unicast/Multicast bits. They are observable and impermanent.



Mechanics (Function)

Switches passively learn source MACs and forward to destination MACs. Routers strictly rewrite MACs at network boundaries.



Security (Fallibility)

Identifiers are not authenticators. Robust network security demands 802.1X authentication and a defense-in-depth approach.